

GFAz: Order-of-Magnitude Pangenome Analytics by Computing over Compression

84×
Compression ratio
369 GB → 4.5 GB

1.36 GiB/s
Compression throughput
CPU · 32 threads

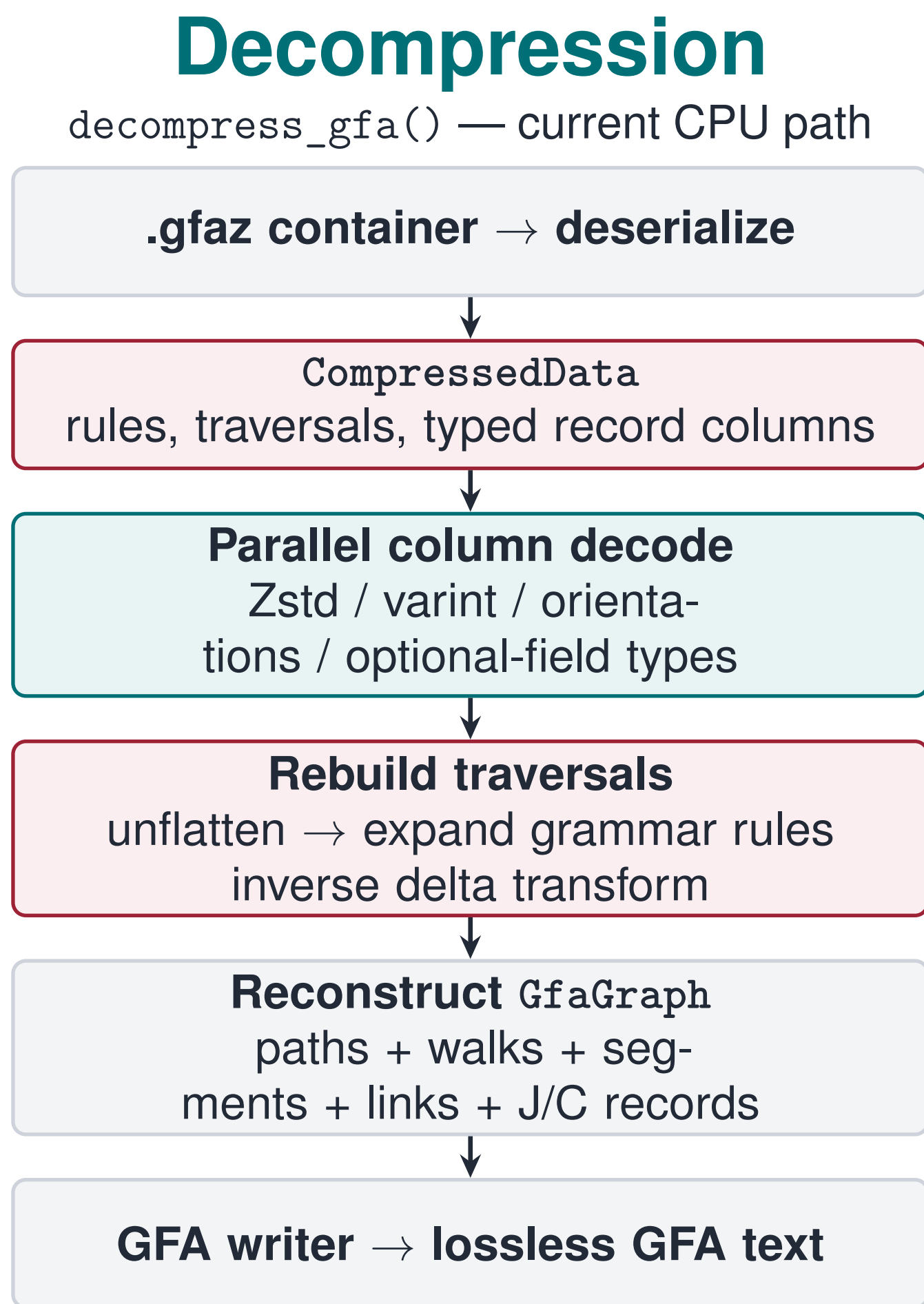
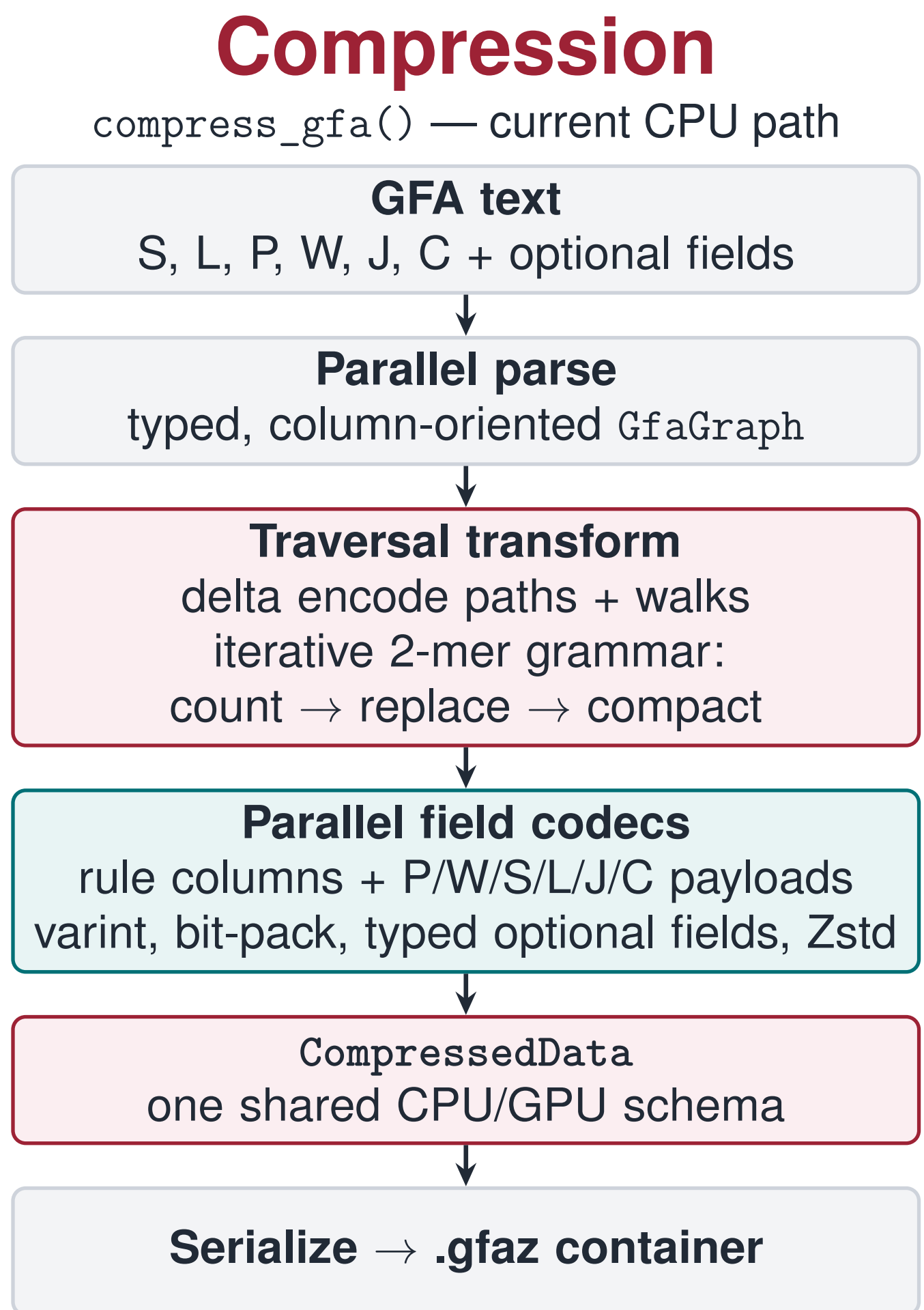
5.43 GiB/s
Decompression throughput
CPU · 32 threads

619×
Compute speedup
up to

25×
Lower compute memory
up to

1 Compression engine

lossless · full GFA semantics · 32-core CPU path

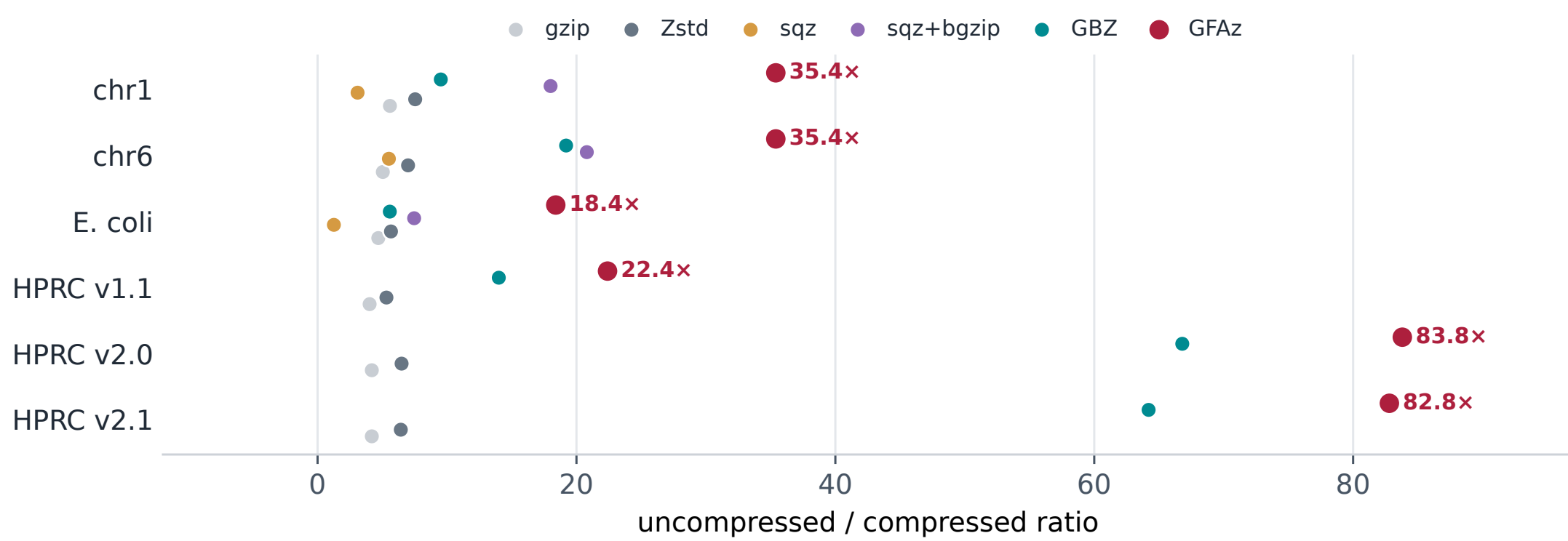


Key design: a recursive grammar built from native 2-mer rules

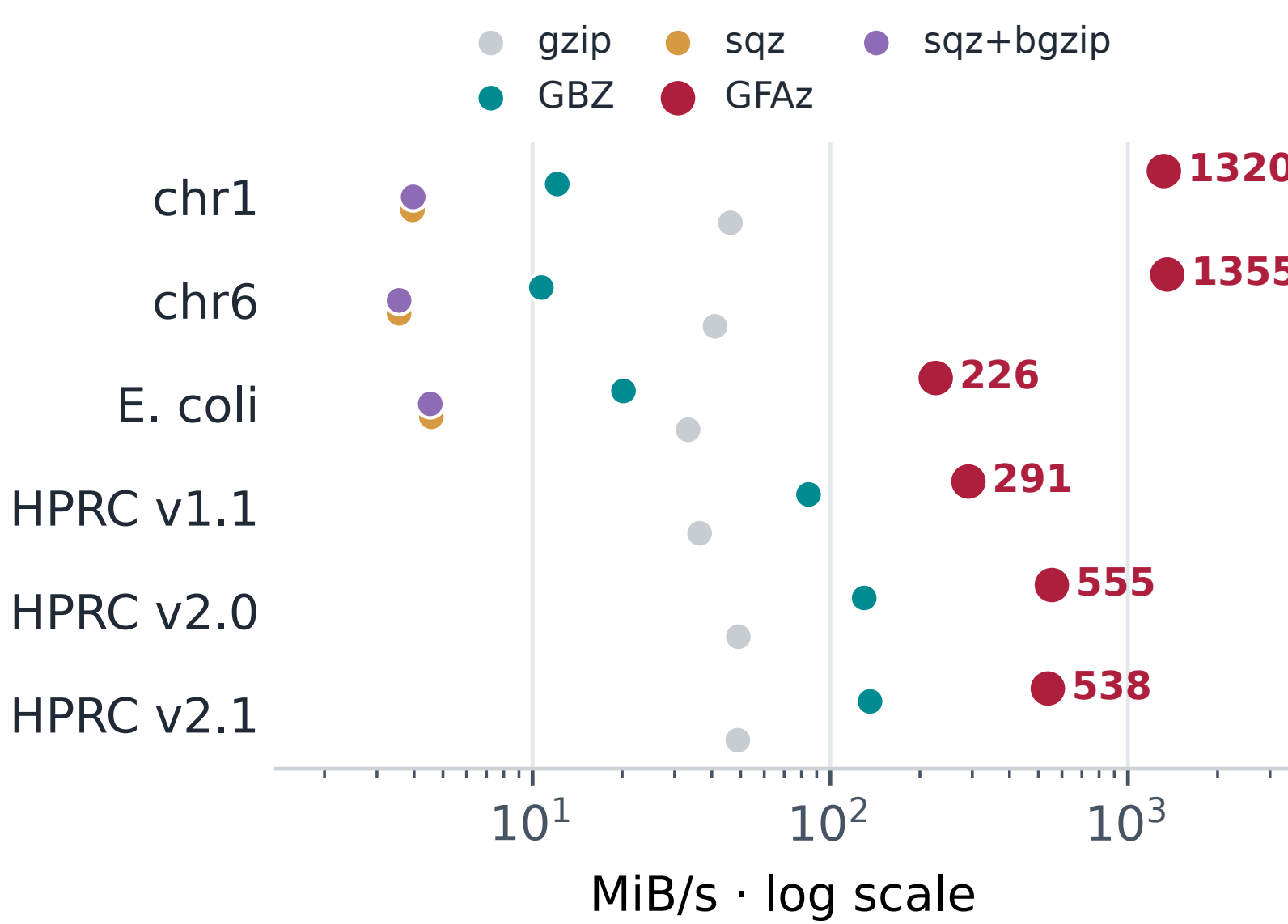
Input	—	12 7 12 7 12 7 12 7 31
Round 1	$R_1 \leftarrow (12, 7)$	$R_1 \ R_1 \ R_1 \ R_1 \ 31$
Round 2	$R_2 \leftarrow (R_1, R_1)$	$R_2 \ R_2 \ 31$
Round 3	$R_3 \leftarrow (R_2, R_2)$	$R_3 \ 31$

Every rule combines exactly **two current symbols**, while later rounds reuse earlier rules to capture longer repeats. Paths and walks share the rulebook. Each round is a count-and-rewrite scan: *O(N)* per round; fixed rounds are linear.

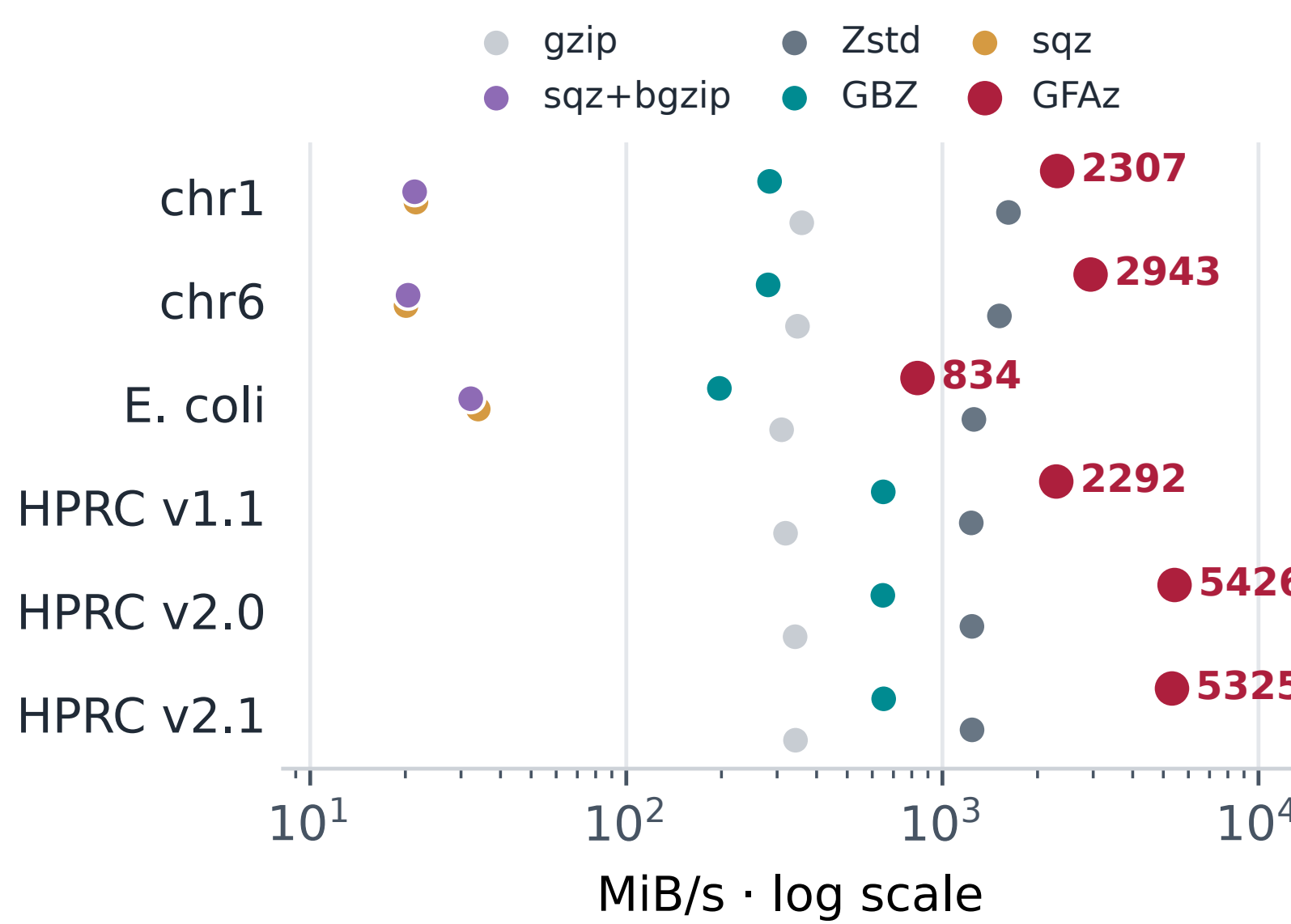
Compression ratio



Compression throughput



Decompression throughput

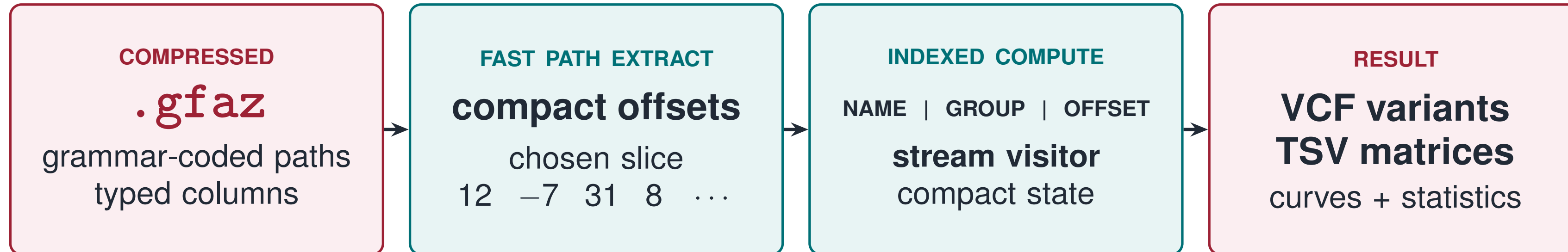


Current README results, CPU only. Ratio is uncompressed size / compressed size; higher is better. GFAz uses 32 threads, and compression is end-to-end CLI wall time. **GFAz reaches 1.36 GiB/s compression and 5.4 GiB/s streaming decompression while retaining the best ratio on every graph.**

2 Compute engine

run the analysis on the compressed container

From compressed paths to analysis results



LOW MEMORY one selected stream + bounded decoder scratch + compact analysis state

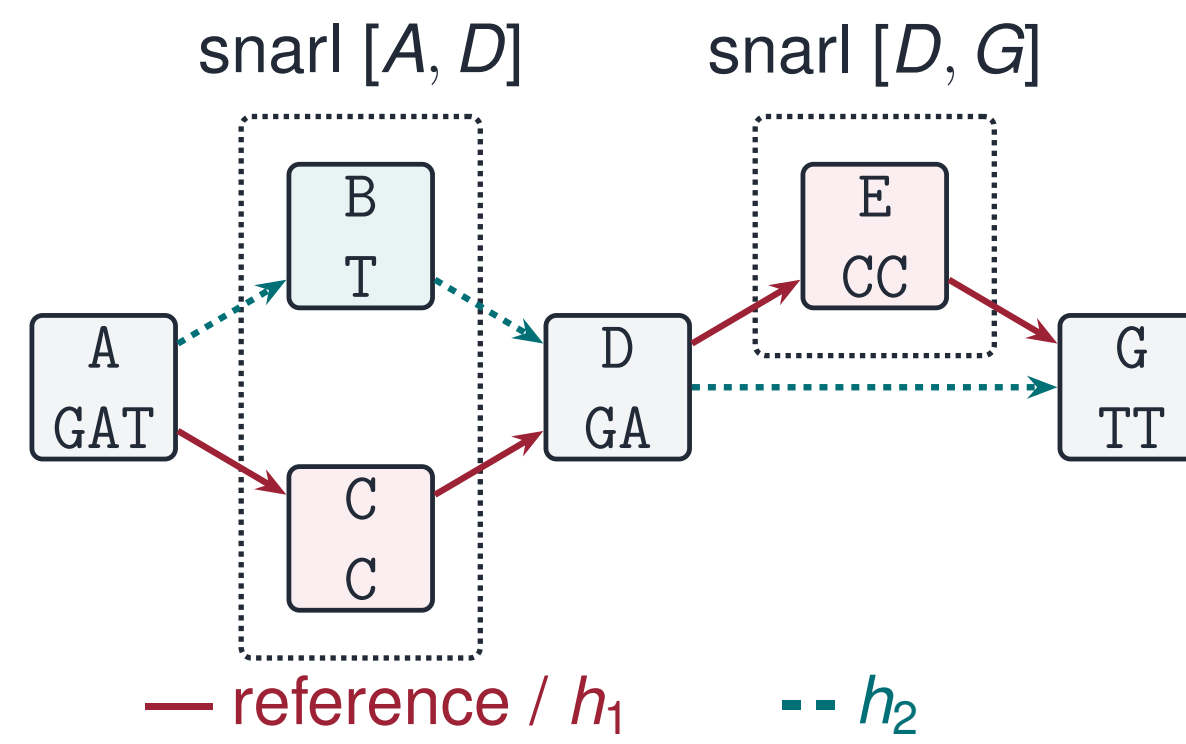
UNUSED STREAMS REMAIN COMPRESSED · THE FULL GFA IS NEVER MATERIALIZED

StreamDecodedNodes expands a chosen slice and immediately invokes the analysis visitor for each signed node. **Only streams used by the query are expanded**; they are consumed online rather than retained as decoded paths.

GFAz command	Reproduces	Passes	Accumulator
deconstruct	vg deconstruct	1 (+topo)	per-site allele table
growth	Panacus	1	coverage → histogram
depth	odgi depth	1	per-node counters
stats	odgi stats	0	metadata totals
pav	odgi pav	3	node→groups CSR
similarity	odgi similarity	2 (+join)	node→groups CSR

Six analyses, one decoder — they differ only in the accumulator. A new analysis is a new visitor.

deconstruct: GFA → VCF with no graph



#CHROM	POS	REF	ALT	INFO	h1	h2
chr1	4	C	T	AC=1;AN=2;AF=0.5	0	1
chr1	6	ACC	A	AC=1;AN=2;AF=0.5	0	1

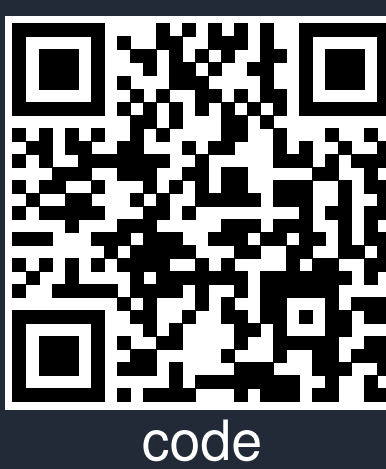
Phase 1 recovers snarls from *topology alone* (biconnected decomposition, $O(V+E)$, decodes zero haplotypes). **Phase 2** streams each haplotype once through a boundary state machine; a reverse boundary pair normalizes inversions. No graph, no general snarl index.

Analysis	Graph	Baseline	GFAz	Speedup	Mem.
deconstruct vs. vg	chr1	805 s / 30.3 GB	11.3 s / 8.3 GB	71×	3.7×
	HGSVC3	132.9 min / 397 GB	8.7 min / 51.9 GB	15×	7.6×
growth vs. Panacus	chr1	18.3 s / 6.16 GB	0.74 s / 0.66 GB	25×	9.3×
	v2.0	245 min / 327 GB	39.8 s / 12.9 GB	369×	25×
pav vs. odgi	chr1	3499 s / 31.9 GB	11.7 s / 9.5 GB	299×	3.4×
	chr6	4759 s / 19.4 GB	7.8 s / 6.4 GB	619×	3.0×

Wall-clock / peak RSS at 16 threads. **Outputs match the baselines**: growth reproduces Panacus' curve exactly, deconstruct agrees with vg to within 0.2% of records, pav matrices are structurally identical to odgi's. On the 358/369 GB HPRC graphs vg and odgi *cannot load the input at all* — GFAz runs from a 4.5 GB container.

What we built with GFAz

- A **state-of-the-art lossless GFA compressor**: 84× smaller, with 1.36/5.43 GiB/s CPU compression/decompression.
- A **compressed-graph compute engine** that uses compact indices and expands only selected path streams—never the full GFA.
- Six exact pangenome analyses** running up to **619×** faster with up to **25×** less memory.



code

Open source: github.com/babypylutokurt/GFAz · Contact: taolue.yang@temple.edu

Paper: Yang, Liu, Jiang, Shi, Jin. *GFAz: Order-of-Magnitude Pangenome Analytics by Computing over Compression*. ICS '26. doi:10.1145/3797905.3807870

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